

PROFILE

Senior Machine Learning Engineer with 8+ years of experience building and deploying ML systems at scale across NLP, LLMs, recommender systems, and computer vision. Strong background in LLM fine-tuning, evaluation, and integration into production workflows, with deep expertise in MLOps (AWS, CI/CD, MLFlow). Proven track record of end-to-end ownership from research to deployment, mentoring engineers, and collaborating with researchers to deliver cutting-edge models into real-world systems.

SKILLS AND COMPETENCIES

Large Language Models (LLMs), Deep Learning (DL), Machine Learning (ML): Huggingface, Transformers, PyTorch, Accelerate, PEFT, AutoGluon, AWS SageMaker, fastai, Ax, Optuna, spaCy.

MLOps and Infrastructure: MLFlow, Weights & Biases, LightningAI, Neptune, Docker, Kubernetes, AWS, GCP, Azure, Prometheus, Grafana, Ansible, CI/CD pipelines.

Data Engineering: MongoDB, Spark, Airflow, Redis, Kafka, Hive, Hadoop.

Full-Stack Development and Deployment: fastAPI, OpenAPI, REST APIs, microservices for inference, Streamlit, D3.js, Git, Test Driven Development (TDD), Front-end frameworks, Back-end frameworks.

Collaboration and Leadership: Mentorship, Code reviews, Agile/Scrum, Research-to-production translation.

EXPERIENCE

R&D Scientist II - Quantum NLP (QNLP)

Quantinuum Ltd.

Oct 2023 — Present

Oxford, United Kingdom

- **LLM for String Diagram Generation and Rewriting**

- Created data generation and GRPO-based model finetuning pipelines for training LLMs for generating and rewriting string diagrams.

- **End-to-End Generation of String Diagrams from Sentences**

- Designed, implemented, and trained a custom BERT-based model for converting sentences directly to string diagrams which are the starting point of QNLP training pipelines . This removed the grammar-based parsing that often required a large amount of expertise to debug.
- Created the dataset used for training the said model and implemented an appropriate data augmentation strategy for the task. This entailed implementing a compact tree-based representation of string diagrams.
- Performed LoRA fine-tuning on the custom model for more efficient training, coupled with automated hyperparameter tuning using Ax. Tracked the experiment results such as metrics and artifacts using both Neptune and MLFlow.
- Mentored interns who worked on this project and worked with them closely so they can contribute to the new features being developed.

- **DisCoCirc Pipeline Implementation**

- Helped with an improved implementation of this paper  which described a model for how texts derive meaning from their constituent sentences.
- Created benchmarking datasets to compare the performance of the new pipeline to an older, different pipeline .
- Wrote reusable, configurable scripts to check the coverage of the new implementation on the dataset. This simplified the testing pipeline and increased the confidence with the new pipeline. Wrote about these results in an accompanying preprint.
- Implemented visualization functions to allow inspection of the output circuit diagrams.
- Presented work to other colleagues not necessarily familiar with QNLP.

- **lambeq Research and Development**

- Contributed to the **lambeq open-source library**, implementing new features and bug fixes, enhancing APIs, and improving efficiency for downstream ML research tasks.
- Mentored interns and conducted code reviews, raising the overall code quality and maintainability of the library.

Data Science Engineer

AI Cross Inc.

Nov 2021 — Sep 2023

Tokyo, Japan

- **Multiple ML-based Platform-as-a-Service (PaaS) Products**

- Designed and built a ML PaaS product  within the AWS ecosystem, enabling customers to **train and deploy models via APIs** at scale.
- Designed and built a Business Marketing PaaS product  using the AWS ecosystem, allowing customers to **perform A/B testing and optimize their marketing strategies** with tight integration into their user-facing applications.
- Delivered prototypes for the above services under tight deadlines (1-2 months), directly contributing to product pitches to and adoption by clients.
- Partnered with researchers, engineers, and product teams to align ML solutions with user needs and fast-track development of new features.

Machine Learning Engineer

LINE Fukuoka Corporation

Sep 2019 — Nov 2021

Fukuoka, Japan

- **Detection of High-Profile Individuals from Caricatures**

- Developed **large-scale image recognition pipeline (1M+/s throughput)** based on the WarpGAN paper with Spark and Airflow.
- Built APIs for real-time inference and visualization of batch detection results; added **CI/CD pipeline and monitoring** with Prometheus and Grafana.

- **Content Suggestion via Image-Based Emotion Recognition**

- Collaborated with researchers and product team to design and deploy an emotion detection-based recommendation system as a new feature to the LINE messaging application.

- **Machine Learning Tooling and Knowledge-Sharing Tasks**

- Improved existing and created new ML/MLOps tools with the team.
- Gave internal talks on the following areas of interest: meta-learning, model quantization , auto-ML , causal inference , secure AI , and explainable AI .

Data Scientist

Teradata Corporation

Nov 2016 — Sep 2019

Manila, Philippines

- **Survey Response Classification**

- Built NLP models (LSTM + embeddings) for **classification of user responses** for automotive predictive maintenance, improving automation.
- Designed an **end-to-end ML system** with SQL ingestion pipelines, Dockerized deployments, and visualization apps.
- Conducted knowledge-sharing sessions on ML advances for the wider team.

Senior Full-Stack Engineer

icannhas Inc.

Sep 2011 — Nov 2016

Manila, Philippines

- Developed scalable full-stack solutions with API-driven architectures and predictive features.

EDUCATION

Master of Science in Physics

University of the Philippines Diliman, Quezon City (On leave)

2021 — Present

Bachelor of Science in Civil Engineering, magna cum laude

University of the Philippines Diliman, Quezon City

2006 — 2011

Awardee, Engineering Top 100 (out of 5000 students) in Academics, 2011

10th Place, ACM International Collegiate Programming Contest Asia Manila Regional Contest, 2009

Honorable Mention, 37th International Physics Olympiad, 2006